

Assignment 1

Q.1.

Blockchain, in simple terms, is a distributed database that everyone can get a copy of. Every person with a copy can add new records to this database but cannot change any record that is already present in there. This property makes a blockchain great to record data in a transparent way because everyone gets to see what's in it. Its use cases can be seen as:

- a. Cryptocurrencies : It allows people to directly transact with one another without having to trust third parties like banks. Over 1600 different cryptocurrencies have been created.
- b. Odometer Fraud: By tampering with the odometer, someone can make a car appear to be newer and less worn out resulting in customers paying more than what the car is actually worth. The government tries to counter that by collecting the mileage of cars when they get a safety inspection but that's not enough.
So, instead we could replace regular odometers with smart ones that are connected to the internet and frequently write the mileage to a blockchain. This would create a secure and digital certificate for each car. And because we use a blockchain, no one can tamper with data and everyone can look at a vehicle's history.
- c. Notary : A notary is someone who can confirm and verify signatures on legal documents. But we can use blockchain for that. For example, the online website stampd.ip allows users to add documents to the Bitcoin or Ethereum Blockchain. Once added, you can always prove that you created a document at a certain point in time much like a notary (although right now blockchains are not on the same level as notaries in legal perspective).
- d. Digital Voting: Right now voting happens either on paper or special computers that are running proprietary software. Voting on paper costs a lot of money and electronic voting has security issues. But instead of these, we could use blockchains to cast and store votes. Such a system would be transparent.
The Swiss Company Agora is already working on such a system and it's going to be completely open source. But it has many challenges. First we have to be able to identify voters without compromising their privacy. Secondly if we allow people to vote with their own computers or phone, we must take into account that those might be infected with malware designed to tamper with the voting process.
- e. As above 4 points, blockchains can be used to keep track of information and verify its integrity. But blockchain can be more powerful by adding smart contracts to them. Smart contracts are tiny computer programs that live on

blockchain and can perform actions when certain conditions are met.

Insurance companies can use smart contracts to validate claims and calculate a payout. With smart contracts we can secure our own data on blockchain.

As far as being relevant to IITK campus community, we can extend the use for academic credential verification. Blockchain can store academic credentials like degrees, transcripts, and certificates securely. Students and alumni could share these credentials with employers or institutions without the need for third-party verification. Platforms like **Blockcerts** offer solutions for blockchain-based credentialing.

Q.2.

a. Bitcoin (BTC)

- **Specific Use Case:**

This is the original plan for BTC—to be used as an electronic peer-to-peer payment system. BTC can be sent from your corner of the world to someone you've never seen before in another continent without third-party interference.

- **Consensus Mechanism:**

Bitcoin uses the algorithm Proof of Work (PoW). The central idea behind this algorithm is to solve a complex mathematical puzzle and easily give out a solution. This mathematical puzzle requires a lot of computational power and thus, the node who solves the puzzle as soon as possible gets to mine the next block.

b. Ethereum (ETH)

- **Specific Use Case:**

One of the main real-world use cases for Ethereum is decentralized finance (DeFi) applications. Here we can find a wide variety of functionalities ranging from decentralized lending (based on smart contracts), decentralized exchanges and the creation of stablecoins.

- **Consensus Mechanism:**

Ethereum's consensus mechanism is Proof of Stake (PoS). In PoS, validators are chosen to validate transactions and create new blocks based on how much ETH they stake. Validators who stake more are more likely to be chosen.

c. Solana (SOL):

- **Specific Use Case:**

The most obvious use case of Solana blockchain is a decentralized application or dApp development. Decentralized applications gained momentum with the arrival of smart contract programmability on Ethereum. Solana also supports the development of smart contracts, which makes it a suitable candidate for dApp development.

Consensus Mechanism:

Solana's consensus mechanism is a hybrid of Proof of History (PoH) and Proof of Stake (PoS):

Proof of History (PoH)- A time-stamping machine, constantly recording the passage of time on the blockchain. This allows participants to agree on the order of events without needing a constant communication medium, freeing up the dedicated resources used for communication thus enhancing the transaction processing speed.

Proof of Stake (PoS)- A consensus mechanism where validators stake their cryptocurrency as collateral to earn rewards for helping the blockchain.

d. Hyperledger Fabric:**Specific Use Case:**

One of the most common use cases for Hyperledger Fabric is supply chain management. The technology allows for greater transparency and traceability throughout the supply chain, making it easier to track goods and materials from the point of origin to the point of consumption.

Consensus Mechanism:

Hyperledger Fabric offers multiple consensus mechanisms. Some of the most used mechanisms are : 1) Practical Byzantine Fault Tolerance (PBFT) : A consensus algorithm that provides fault tolerance and reliability in networks with a limited number of trusted participants. 2) RAFT: A consensus algorithm that maintains a consistent state across multiple nodes, and is well-suited for networks with unknown or potentially untrusted participants . 3) Solo: A consensus algorithm used for testing purposes in a single-node network, but not suitable for production use.

e. Corda :**Specific Use Case:**

Corda accelerates and automates settlement processes, thereby removing the need for costly and risky reconciliation processes, reducing credit, and counterparty risk and default risk arising from unsettled trades.

Consensus Mechanism:

R3 Corda's consensus mechanism is called "Pluggable Consensus", which allows participants to choose a consensus algorithm that best suits their business needs.

Q.3.

I used Python to write this and checked the answer on an online compiler(named OneCompiler) . First I tried C++ but that required openssl installation and a few more for its execution which I was unable to pull off.

```

import hashlib
import time

def sha256_hash(text):
    return hashlib.sha256(text.encode()).hexdigest()

def find_nonce(input_string):
    threshold = '000ff'
    nonce = 0
    start_time = time.time()

    while True:
        if sha256_hash(input_string + str(nonce))[5] == threshold:
            return nonce, time.time() - start_time
        nonce += 1

def main():
    input_string = input("Enter the string: ")
    nonce, time_taken = find_nonce(input_string)
    print(f"Nonce: {nonce}\nTime taken: {time_taken:.4f} seconds")

if __name__ == "__main__":
    main()

```

In this, I set the threshold as **000ff**. This means the program looks for a hash that starts with "000".

And when I chose the string “**BlockBloom**” as input in above code I got output:

```

Nonce: 541113
Time taken: 0.6916 seconds

```

The time taken was varying approx between 0.59 seconds to 0.71 seconds after repeated executions.

Q.4.

An unspent transaction output (UTXO) is the amount of digital currency that remains after a cryptocurrency transaction. You can think of it as the change you receive after buying an item, but it is not a lower denomination of the currency—it is a transaction

output in the database generated by the network to allow for non-exact change transactions.

Q.5.

A blockchain is called immutable because it uses cryptographic hashing to lock in the data stored in each block. When a block is created, it contains a unique hash, which is like a digital fingerprint of its data. Each block also includes the hash of the previous block, creating a chain of linked blocks.

If someone tries to change the data in a block, its hash will change, breaking the connection to the next block. The network will immediately notice this mismatch, making it clear that something has been tampered with. Because every block is connected, changing one block would require altering all subsequent blocks, which is practically impossible due to the computing power needed. This design ensures that once data is added to the blockchain, it remains secure, permanent, and unchangeable, providing a trustworthy record of transactions.

Q.6.

When a criminal adds a fraudulent block to a PoW blockchain without having the power to execute a 51% attack, the network automatically resolves the fork by favoring the longer, "honest" chain maintained by the majority of miners. The fraudulent block is rejected as an "orphan block," and the network continues to build on the legitimate chain. This happens because the blockchain prioritizes the chain with the highest cumulative computational work (proof-of-work).

Q.7.

In Proof-of-Stake (PoS) systems, the "nothing-at-stake" problem happens when validators can support multiple competing forks of the blockchain at the same time without any real cost. This can cause confusion and allow bad actors to double-spend tokens on different chains. To prevent this, PoS systems use penalties like "slashing," where validators lose some of their staked coins if they validate blocks on conflicting forks.

Q.8.

A 51% attack is less likely with Proof of Stake (PoS) than with Proof of Work (PoW) because it's more difficult and expensive for an attacker to obtain the required cryptocurrency or computing power. In a Proof-of-Stake (PoS) system, an attacker would need to control 51% of the staked cryptocurrency to compromise the network. This is highly improbable because the ownership of the blockchain is widely distributed among many users. Additionally, an attacker would risk losing the entire value of their stake due to the potential depreciation of the cryptocurrency caused by their attack. In contrast, in a Proof-of-Work (PoW) system, an attacker would need to control more than 50% of the network's mining power to launch an attack. This is

extremely challenging and costly, as it requires vast amounts of computational resources.

Q.9.

In blockchain technology, a digital signature is a cryptographic method used to confirm the validity and integrity of a transaction or data within the network. It functions as a secure, electronic "signature" that verifies the sender's identity and ensures the data has not been altered. This enables trust in the decentralized system without relying on a central authority. Digital signatures are generated using a private key and can be validated using the associated public key, ensuring that the sender cannot deny their involvement (non-repudiation) and safeguarding the data from unauthorized changes.

Q.10.

Oracles are third-party services that fetch and supply data from external sources to the blockchain. They act as the "bridge" between on-chain (blockchain) and off-chain (external) systems. The "oracle problem" refers to the possibility that trusted oracles could be hacked, disseminating inaccurate data over a blockchain. The oracle problem introduces a degree of error into any application in the real world, although the degree of that error varies depending on the application.

Chainlink uses a decentralized network of oracles to fetch, verify, and deliver off-chain data to smart contracts on the blockchain. Chainlink ensures reliable data through decentralized feeds, aggregation contracts for consensus, and a reputation system to track oracle performance. Cryptographic proofs verify data integrity, and staking LINK tokens incentivizes honest oracle behavior by penalizing malicious actions.

Q.11.

Zero-Knowledge Proofs (ZKPs) are a cryptographic method that allows one party to prove they know certain information without revealing the actual details of that information. In a Zero-Knowledge Proof, there are two parties: the prover and the verifier. The prover demonstrates they possess specific knowledge (such as a secret or credential) by performing a series of actions that can only be done if they know the required information. The verifier asks questions to the prover, and if the prover is wrong, the verifier can detect it with a high probability, ensuring that the prover indeed knows the secret.

In the context of blockchains, ZKP's uses are:

Enhancing Privacy: ZKPs enable users to prove ownership of certain data or credentials without disclosing the actual data itself. This preserves privacy, which is critical because blockchain networks can sometimes be vulnerable to data leaks.

Ensuring Transaction Integrity: ZKPs are used to confirm the validity and security of transactions. This is especially important for regulated industries, like banking, where transaction integrity is essential to comply with legal standards.

Enabling Secure Identity Sharing: ZKPs allow users to share their identity securely on blockchain platforms, ensuring that sensitive information remains private while still proving authenticity.